

4. The following table shows the composition of the atmosphere.

Gas	Chemical formula	Percentage found in the atmosphere (%)
argon	Ar	0.93
carbon dioxide	CO ₂	0.0360
helium	He	0.0005
hydrogen	H ₂	0.00005
methane	CH ₄	0.00017
neon	Ne	0.0018
nitrogen	N ₂	78.08
nitrous oxide	N ₂ O	0.00003
oxygen	O ₂	20.95
ozone	O ₃	0.000004

Use the table to answer parts (a) and (b).

(a) (i) Name **two** gases that occur as single atoms. [1]

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(ii) Name **two** elements that occur as molecules. [1]

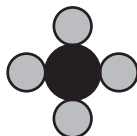
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(iii) Name the gas that has the **lowest** percentage. [1]

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- (b) (i) If  represents a carbon atom, name the gases from the table that can be represented by the following diagrams. [2]



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- (ii) Use the information in part (i) to draw the diagram for oxygen gas. [1]

- (c) The atmosphere can become polluted by substances known as chlorofluorocarbons (CFCs). One example of a CFC is CH_2ClF .

- (i) State how many hydrogen atoms are present in CH_2ClF . [1]

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- (ii) Give the **total** number of atoms shown in the formula, CH_2ClF . [1]

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